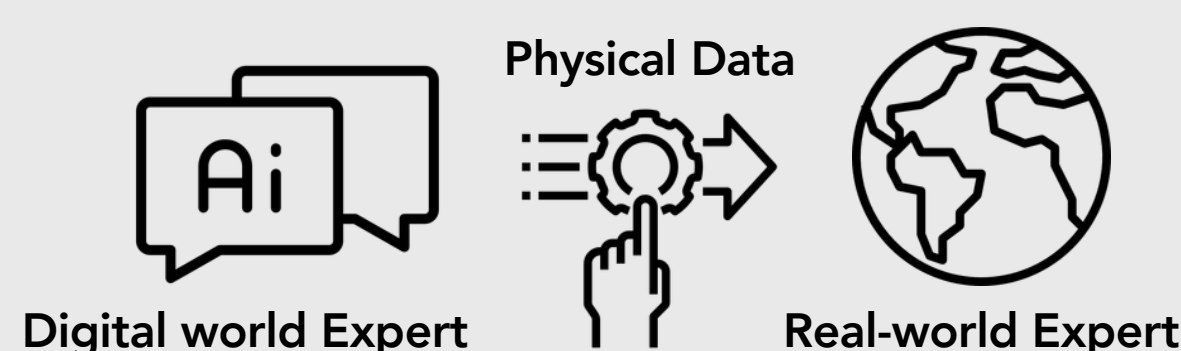


THE CONTEXT & KNOWLEDGE

1. PROBLEM & MOTIVATION

Vision-Language Model (VLM) has demonstrated a strong capability in understanding both visual and text input. This benefits the software automation such as zero-shot ORC and Document summarizer system. Expanding these capabilities of VLM to physical world will allow more complex use cases to the industrial automation. However, the study in Vision-Language-Action (VLA) still very narrow compared to the VLM due to several constraints including hardware and computational power requirement. In this project, we propose an End-to-End framework for develop VLA model on the consumer grade setup enable VLA research more reachable worldwide.



2. DESIGN CONSTRAINTS

- Limited dataset** quantity 1 hour of motor operation covering three task types (SOTA model used > 10 hours).
- Resource requirement** model must fit within consumer GPU (≤ 8 GB VRAM).
- Safety and inference consistency**, robot joints degree of freedom are fixed to a predefined configuration for Dobot CR5.

3. KEY KNOWLEDGE & COURSES

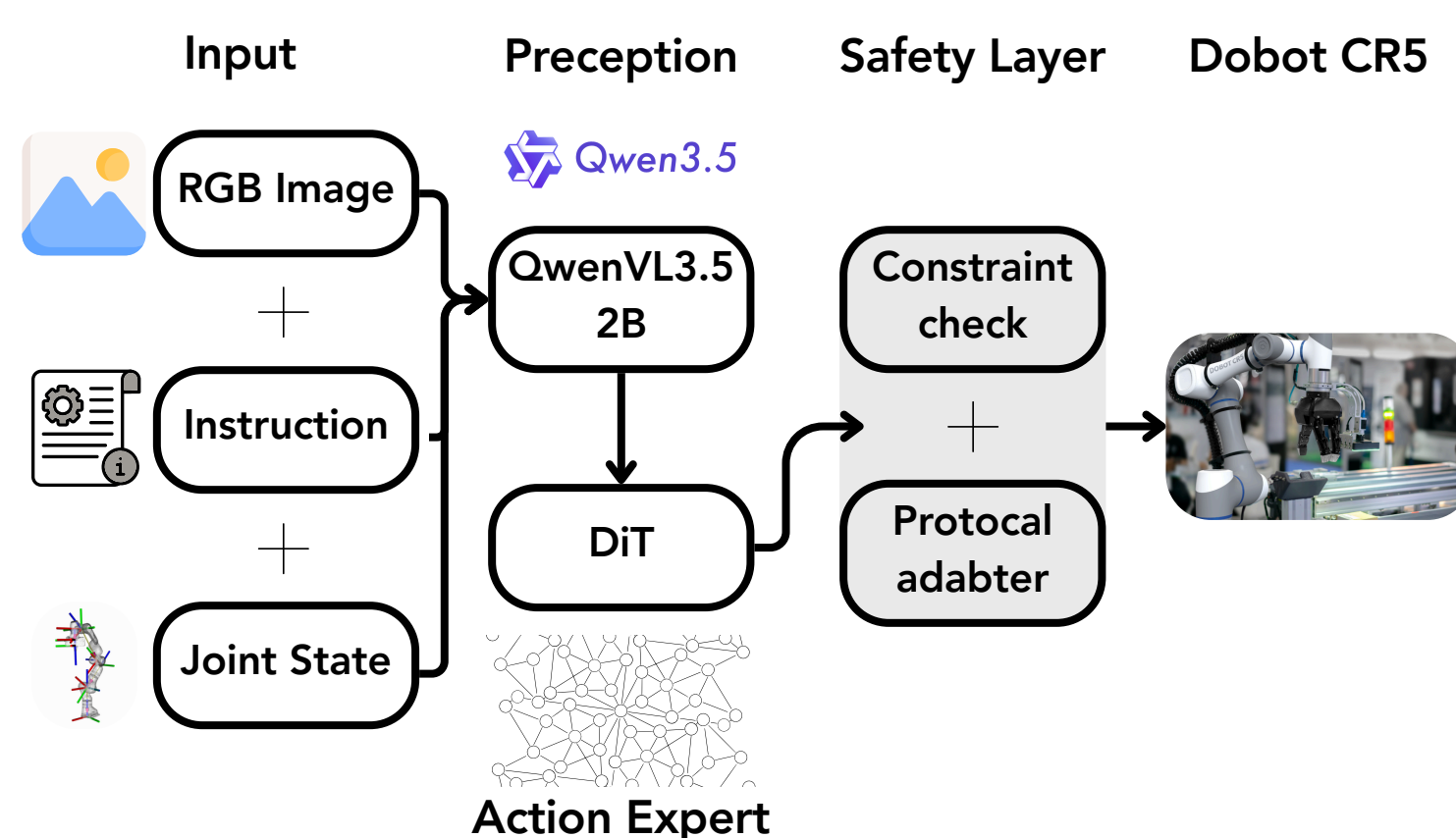
- Kinematics** for robot control and dataset preprocessing and collection
- Mathematics** in calculus, probability, linear algebra to validate AI model architecture
- Data Science** for data-driven base analysis to import the framework and AI model performance
- Deep Learning** for adapting SOTA architecture and training techniques to implement in the project
- Programming** in Python for develop memory efficient training pipeline

4. OBJECTIVES

- VLA model** trained using the proposed framework under 8 GB of VRAM usage with the task success rate >50% (baseline ~28%)
- Framework** for VLA development which enables simulation and real-world robot dataset alignment

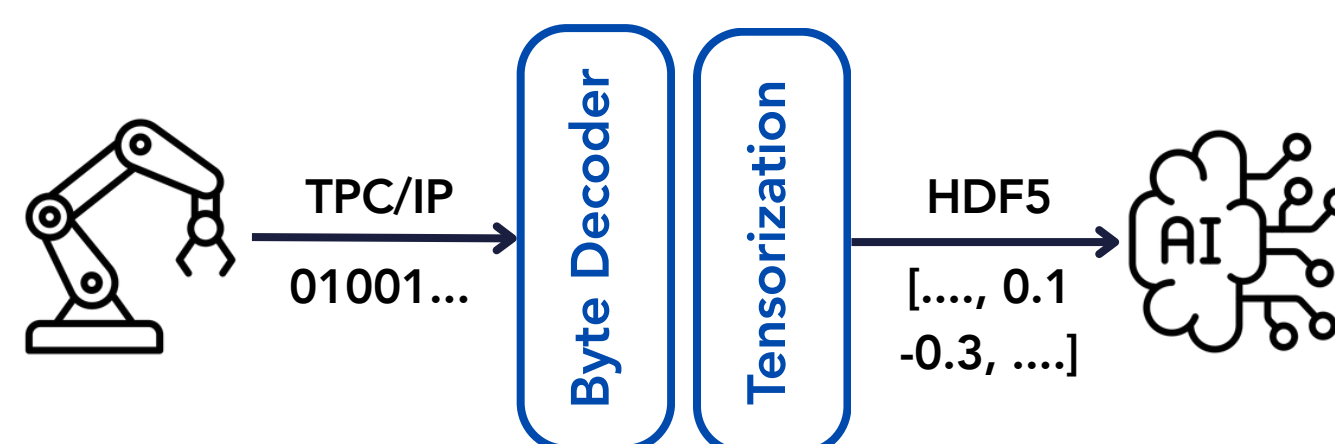
THE CORE DESIGN & IMPLEMENTATION

5. SYSTEM ARCHITECTURE

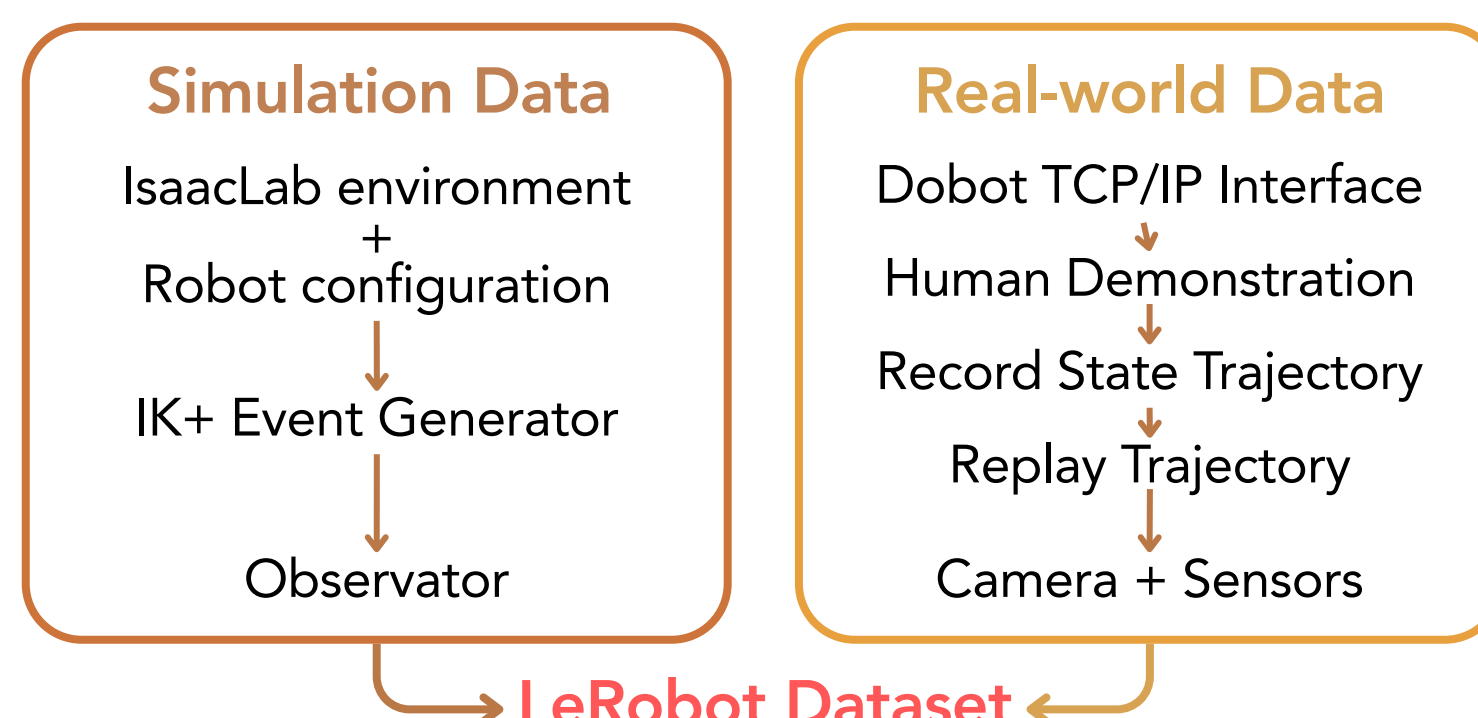


6. IMPLEMENTATION

Phase 1: Robot to AI communication



Phase 2: Dataset Preparation

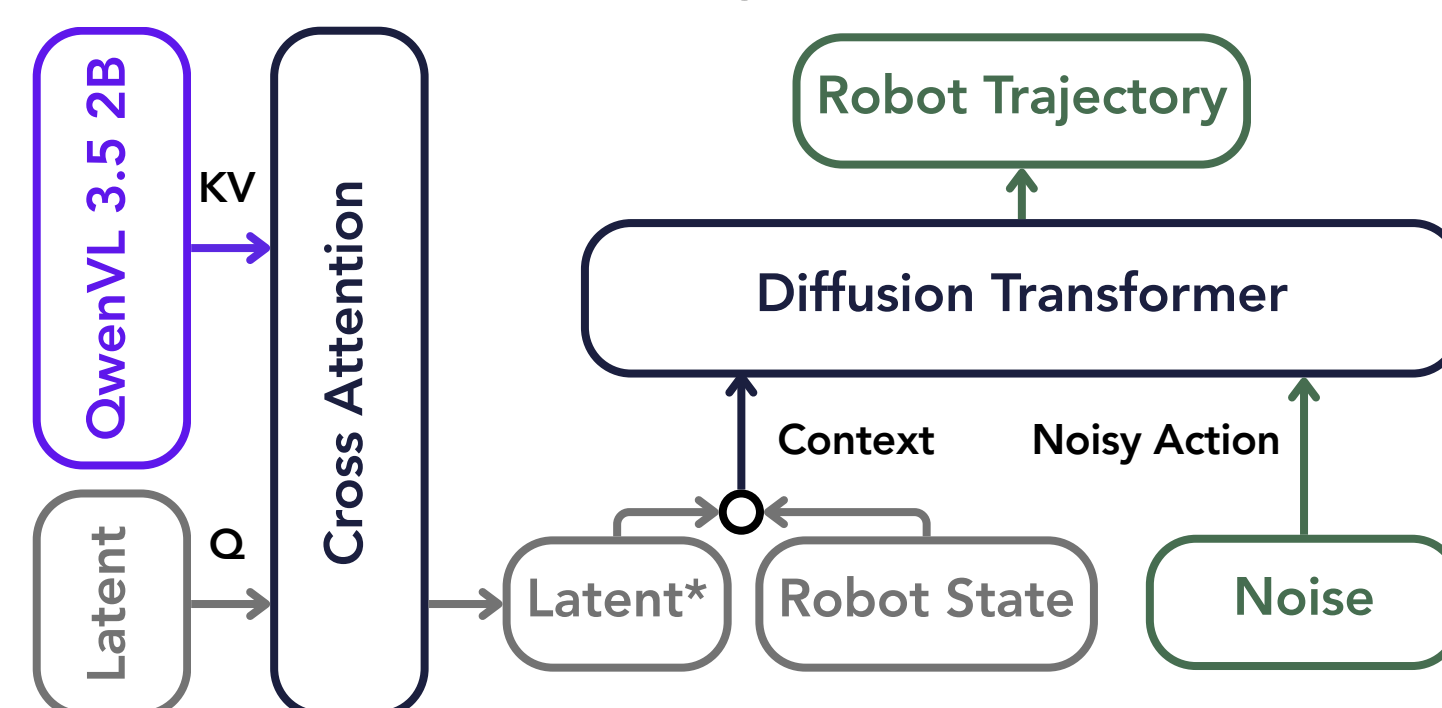


Simulation dataset: 530 episodes with 1 hour long

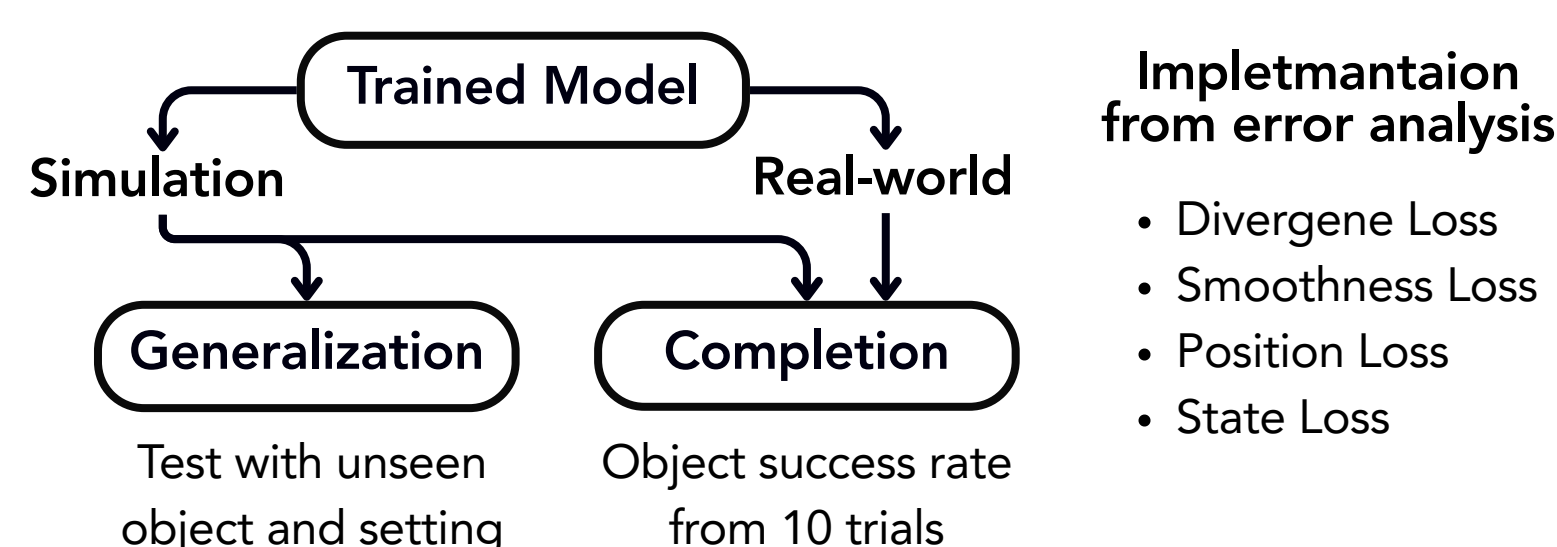
Real dataset: 90 episodes with 10 min long

Task variance: 3 tasks (Stay, Reach and Grip)

Phase 3: Model Training



Phase 4: Model Evaluation and Improvement



Scan for demo video

EVALUATION, DELIVERABLES & IMPACT

7. TESTING & VALIDATION

Simulation - All task combines

	Pass	Fail
Baseline	5	15
Our	11	9

The Baseline model Success rate: 20%

Our model Success rate: 55%

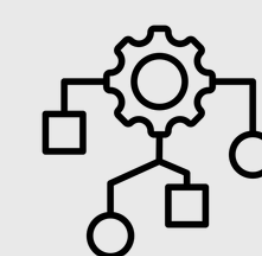
Real World - Reach only

	Pass	Fail
Baseline	0	10
Our	3	7

The Baseline model Success rate: Failed

Our model Success rate: 30%

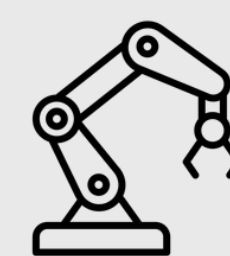
8. DELIVERABLES



End-to-End VLA model development framework on Dobot CR5.



Robotics dataset for training VLA on Dobot CR5 embody.



9. GLOBAL & SOCIAL IMPACT

- Drive robotics open-source community** with our novel framework



huggingface_hub

- The new industrial automation Era VLA** enable robot's robustness and ability to perform more complex task

10. CONCLUSION

This project successfully bridged the gap between high-level semantic instruction and low-level robotic control for the Dobot CR5 platform.

Summary of contribute

- Hardware Accessibility:** Developed a resource-optimize VLA framework capable of inference with only 4GB of vram
- Performance Gain:** Demonstrated a 55% success rate in multi-task sdimulations, more than doubling the baseline performance of 25%

Future Work

- Scale data and task complexity.
- Enhance robustness and failure recovery.